

$\partial L / \partial W_1 = \text{input } X \text{ convolved with } \partial L / \partial Z_1$

x_{11}	x_{12}	x_{13}
x_{21}	x_{22}	x_{23}
x_{31}	x_{32}	x_{33}

input X , 3×3

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$\partial L / \partial z_{11}$	$\partial L / \partial z_{12}$
$\partial L / \partial z_{21}$	$\partial L / \partial z_{22}$

$\partial L / \partial Z_1$, 2×2

=

$\partial L / \partial w_{11}$	$\partial L / \partial w_{12}$
$\partial L / \partial w_{21}$	$\partial L / \partial w_{22}$

$\partial L / \partial W_1$, 2×2

top-left entry: lay $\partial L / \partial Z_1$ on the highlighted window of X , multiply cell by cell, add:

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial z_{11}} x_{11} + \frac{\partial L}{\partial z_{12}} x_{12} + \frac{\partial L}{\partial z_{21}} x_{21} + \frac{\partial L}{\partial z_{22}} x_{22}$$